



DO-IT-YOURSELF: PROJECT

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

```
sketch_nov07d $  
  
#include <Wire.h>  
  
#include "MAX30105.h"  
  
#include "heartRate.h"  
#include <Arduino.h>  
#if defined(ESP8266)  
  /* ESP8266 Dependencies */  
  #include <ESP8266WiFi.h>  
  #include <ESPAsyncTCP.h>  
  #include <ESPAsyncWebServer.h>  
#elif defined(ESP32)  
  /* ESP32 Dependencies */  
  #include <WiFi.h>  
  #include <AsyncTCP.h>  
  #include <ESPAsyncWebServer.h>  
#endif  
#include <ESPDash.h>  
  
/* Your WiFi Credentials */  
const char* ssid = "Jio-Fiber-4G"; // SSID  
const char* password = "1234567890"; // Password  
  
/* Start Webserver */  
AsyncWebServer server(80);  
  
/* Attach ESP-DASH to AsyncWebServer */  
ESPDash dashboard(server);
```

Fig. 3: Code for setting the Wi-Fi SSID and ESP server

```
void loop()  
{  
  long irValue = particleSensor.getIR();  
  
  if (checkForBeat(irValue) == true)  
  {  
    //We sensed a beat!  
    long delta = millis() - lastBeat;  
    lastBeat = millis();  
  
    beatsPerMinute = 60 / (delta / 1000.0);  
  
    if (beatsPerMinute < 255 && beatsPerMinute > 20)  
    {  
      rates[ratesSpot++] = (byte)beatsPerMinute; //Store this reading in the array  
      ratesSpot %= RATE_SIZE; //Wrap variable  
  
      //Take average of readings  
      beatAvg = 0;  
      for (byte x = 0; x < RATE_SIZE; x++)  
        beatAvg += rates[x];  
      beatAvg /= RATE_SIZE;  
    }  
  }  
  float temperature = particleSensor.readTemperature();  
  
  Serial.print("temperatureC=");  
  Serial.print(temperature, 4);  
  
  Serial.print("IR=");  
  Serial.print(irValue);  
  Serial.print(", BPM=");  
  Serial.print(beatsPerMinute);  
  Serial.print(", Avg BPM=");  
  Serial.print(beatAvg);  
  
  Serial.println();  
}
```

Fig. 4: Code snippet for updating the sensor data in web dashboard

parts—the IoT sensor battery charger (left side) and the IoT sensor sticker.

After connecting the components as per circuit diagram, interconnect the charger and sensor circuits. The charging circuit has battery connector JP1, which is connected to the IoT sensor chip circuit via switch S1. Connect the battery's positive terminal through wire to pin 1 of JP1 and the negative terminal through another wire to the battery connector.

Software

You need to prepare the code for IoT sticker and Arduino IDE. To install the Arduino IDE and ESP8266 board, open the library manager and install ESP-DASH and Sparkfun Max301x libraries. Include the Max3015 library and ESP-DASH in the code. Then set up the SSID and password of your

phone Wi-Fi (hot spot) or home Wi-Fi. To be mobile you may use phone Wi-Fi, but if you want to monitor at home only you may use the home Wi-Fi.

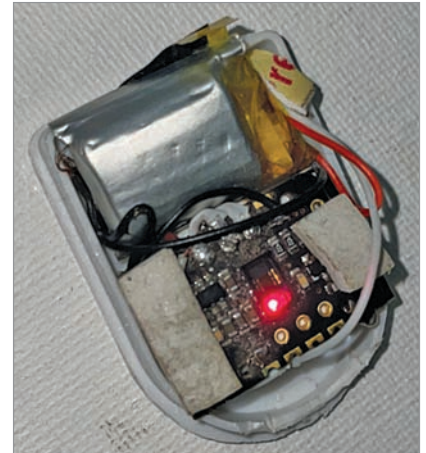


Fig. 5: Author's prototype for the sticker with battery



Fig. 6: The author using the device